

PART A [25 marks]

This section consists of 3 questions. Answer **all** questions.

1 a) Find the value of m if $\lim_{x \rightarrow 0} \frac{mx + 3x^2}{4x - 8x^2} = 3$. [3 marks]

b) Find the horizontal and vertical asymptotes for $f(x) = \frac{3x}{\sqrt{x^2 - 16}}$.

[6 marks]

2 a) Find the derivative of $f(x) = \frac{6}{\sqrt{x}}$ by using the first principle. [5 marks]

b) Find the value of $\frac{dy}{dx}$ if $\frac{1}{y} - \frac{1}{x} = 3$ when $x = \frac{1}{2}$. [6 marks]

3 Given that $f(x) = 3x^4 - 4x^3 + 1$. Use the first derivative test to determine the relative maximum or minimum (if any).

[5 marks]

PART B [75 marks]

This section consists of 7 questions. Answer **all** questions.

1 a) Given $(1 + 4i)u - 4w = -1 + 2i$. Find u and w if u and w are complex numbers with w is a conjugate of u .

[5 marks]

- b) Given $z = x + yi$, where x and y are real numbers. If $\left| \frac{z+i}{z+1+i} \right| = 2$, show that
- $$3x^2 + 3y^2 + 8x + 6y + 7 = 0.$$
- [5 marks]**
- 2 a) Solve the equation $\ln x - \frac{3}{\ln x} = -2$. **[5 marks]**
- b) Find the solution set of $|2 - 3x| < |x + 3|$. **[6 marks]**
- 3 a) A function $f(x)$ is defined by $f(x) = \frac{3x}{x-6}$ for $x \neq 6$. Find the values of x such that $(f \circ f)(x) = 0$. **[4 marks]**
- b) Given $f(x) = \sqrt{1-3x}$ and $g(x) = \frac{x}{2} - 1$. Find $f\left(g^{-1}\left(-\frac{7}{2}\right)\right)$. **[4 marks]**
4. Given $f(x) = \ln(2x+3)$ and $g(x) = \frac{e^x - 3}{2}$.
- a) Find $(f \circ g)(x)$ and $(g \circ f)(x)$. Hence, state the conclusion about the results. **[5 marks]**
- b) Sketch the graphs of $f(x)$ and $g(x)$ on the same axes. Hence, State the domain and range of $f(x)$. **[5 marks]**
- 5 a) Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{x+2} - \sqrt{2}}{x}$. **[3 marks]**

b) Given that

$$f(x) = \begin{cases} 1 + e^x & , \quad x \leq 0 \\ \frac{x+6}{3-x} & , \quad 0 < x \leq 4 \\ C & , \quad x > 4 \end{cases}$$

where C is a constant.

i) Determine whether $f(x)$ is continuous at $x = 0$. **[5 marks]**

ii) Given that $f(x)$ is discontinuous at $x = 4$, determine the values of C .

[3 marks]

6 a) Find $\frac{dy}{dx}$ if $y \ln x = e^{x-y}$. **[4 marks]**

b) Parametric equations of a curve is given by $x = \frac{2t}{t^2 + 1}$ and $y = \frac{3t^2 + 1}{t^2 + 1}$.

Find

i) $\frac{dy}{dx}$ in terms of t . **[6 marks]**

ii) $\frac{d^2y}{dx^2}$ when $t = 1$. **[6 marks]**

7. Water is running at a steady rate of $36\pi \text{ cm}^3 \text{ s}^{-1}$ into a right inverted circular cone with a semi-vertical angle of 45° .

a) Find the rate of increasing in water depth when the water level is 3 cm.

[5 marks]

b) Find the time taken when the depth of the water is 18 cm.

[4 marks]

END OF QUESTION PAPER

SUGGESTED ANSWERS:**PART A**

1 a) $m = 12$

b) H.A. : $y = 3$ and $y = -3$; V.A. : $x = 4$ and $x = -4$

2 a) $f'(x) = -\frac{3}{x^{\frac{3}{2}}}$

b) $\frac{dy}{dx} = \frac{4}{25}$

3) min : $(1, 0)$

SECTION B

1 a) $u = 3 - 2i$, $w = 3 + 2i$

2 a) $x = \frac{1}{e^3}$ or $x = e$

b) $\{x : -\frac{1}{4} < x < \frac{5}{2}\}$

3 a) $x = 0$

b) $f\left(g^{-1}\left(-\frac{7}{2}\right)\right) = 4$

4 a) $(f \circ g)(x) = x$; $(g \circ f)(x) = x$; $g = f^{-1}$ and $f = g^{-1}$

b) $D_f = (-\frac{3}{2}, \infty)$; $R_f = (-\infty, \infty)$

5 a) $\frac{1}{2\sqrt{2}}$

b i) continuous

ii) $C = (-\infty, -10) \cup (-10, \infty)$

6 a) $\frac{dy}{dx} = \frac{e^{x-y} - \frac{y}{x}}{e^{x-y} + \ln x}$

b) i) $\frac{2t}{1-t^2}$

ii) undefined

7 a) $\frac{dh}{dt} = 4 \text{ cm s}^{-1}$

b) $t = 54 \text{ s}$